house price $\quad$ size $\quad$ school $\quad$ # of rms $\quad$ pss

$Y \quad X_1 \quad X_2 \quad X_3 \quad X_4$

Multiple linear regression: Hypothesis test for several parameters

Full model $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \varepsilon_i$ $\quad \varepsilon_i \sim N$

reduce model $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$ $\quad \varepsilon_i \sim N($

$H_0 \quad \beta_3 = \beta_4 = 0$ $\quad$ Vs $\quad$ $H_a \quad \beta_3 \neq 0$ or $\beta_4 \neq 0$

reduced model $\quad$ full model

Setup

Suppose now we wish to test a hypothesis of the form

$$H_0: \beta_p = \beta_{p-1} = \cdots = \beta_{p-k+1} = 0. \quad \text{VS } H_a$$

$p = r+1$

That is, the last k components of the parameter vector $\boldsymbol{\beta}$ are all zero.

$$\beta = [\overbrace{\underbrace{\beta_0\ \beta_1\ \ldots\ \beta_{r-k}}_{\beta_0}}^{p-k \text{ elements}} \ \vdots\ \overbrace{\underbrace{\beta_{r-k+1}\ \ldots\ \beta_r}_{\beta_1}}^{k \text{ elements}}]^T$$

$$X = \left[\begin{array}{ccccc:cccc} 1 & x_{11} & x_{12} & \cdots & x_{1,r-k} & x_{1,r-k+1} & \cdots & x_{1r} \\ 1 & x_{21} & x_{22} & \cdots & x_{2,r-k} & x_{2,r-k+1} & \cdots & x_{2r} \\ \vdots & \vdots & \vdots & & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{n,r-k} & x_{n,r-k+1} & \cdots & x_{nr} \end{array}\right]$$

X_0: $n \times (p-k)$ matrix $\qquad n \times (k)$

To test $(\ )_{k\times 1} = (\ )$

H_0 $\beta_1 = 0$ vs H_a $\beta_1 \neq 0$

we compare

full model $y = X\beta + \varepsilon$

reduce model $y = X_0\beta_0 + \varepsilon$

Let $\boldsymbol{X}_{\mathbf{0}}$ be the matrix containing the first $p - k$ columns of $\boldsymbol{X}$ and let

$$\boldsymbol{\beta}_0 = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{r-k} \end{bmatrix}.$$

$H_0 \ \beta_1 = 0 \quad \text{vs} \quad H_a \ \beta_1 \neq 0$

Observe that H_0 can be expressed equivalently as

$$H_0: \boldsymbol{\eta} = \boldsymbol{X}_0\boldsymbol{\beta}_0 \qquad \text{vs} \quad H_a \ \eta = X\beta$$

Now let

full model
$$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$$
$$\hat{Y} = \widehat{\boldsymbol{\eta}} = \boldsymbol{X}\widehat{\boldsymbol{\beta}}$$
reduced model
$$\widehat{\boldsymbol{\beta}}_0 = (\boldsymbol{X}_0^\top\boldsymbol{X}_0)^{-1}\boldsymbol{X}_0^\top\boldsymbol{y}$$
$$\widehat{\boldsymbol{\eta}}_0 = \boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0$$

$$\underline{y - \hat{\eta}_0} = \underbrace{y - \hat{\eta}}_{①} + \underbrace{\eta - \hat{\eta}_0}_{②}$$

$$(\ ) - (\ ) = (\ ) - (\ ) + (\ ) - (\ )$$

Lemma 8

SSE for reduced model   SSE for full model

$$\underbrace{\sum_{i=1}^{n}(y_i-\hat{\eta}_{0i})^2}_{SSE_R}=\underbrace{\sum_{i=1}^{n}(y_i-\hat{\eta}_i)^2}_{SSE_F}+\underbrace{\sum_{i=1}^{n}(\hat{\eta}_i-\hat{\eta}_{0i})^2}_{SSE_D}.$$

That is,

$$\underbrace{\left\|\boldsymbol{y}-\boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0\right\|^2}_{SSE_R}=\underbrace{\left\|\boldsymbol{y}-\boldsymbol{X}\widehat{\boldsymbol{\beta}}\right\|^2}_{SSE_F}+\boxed{\underbrace{\left\|\boldsymbol{X}\widehat{\boldsymbol{\beta}}-\boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0\right\|^2}_{SSE_D}}.$$

$\|A\|=\sqrt{A^TA}$ $\|A\|^2=A^TA$

Proof of Lemma 8

$$\begin{aligned}
SSE_R &= \|y - X_0\hat{\beta}_0\|^2 \\
&= \|\underbrace{y - X\hat{\beta}} + \underbrace{X\hat{\beta} - X_0\hat{\beta}_0}\|^2 \\
&= \underbrace{\|y - X\hat{\beta}\|^2}_{SSE_F} + \underbrace{\|X\hat{\beta} - X_0\hat{\beta}_0\|^2}_{SSE_D} + \underbrace{2(y - X\hat{\beta})^T(X\hat{\beta} - X_0\hat{\beta}_0)}_{\text{show this equals } 0}
\end{aligned}$$

$$\begin{aligned}
(y - X\hat{\beta})^T(X\hat{\beta} - X_0\hat{\beta}_0) &= \left[y - X(X^TX)^{-1}X^Ty\right]^T(X\hat{\beta} - X_0\hat{\beta}_0) \\
&= \left[(I - \underbrace{X(X^TX)^{-1}X^T}_{H})y\right]^T(X\hat{\beta} - X_0\hat{\beta}_0) \\
&= y^T(I-H)^T(X\hat{\beta} - X_0\hat{\beta}_0) \\
&= y^T\underbrace{(I-H)^TX}_{0}\hat{\beta} - y^T\underbrace{(I-H)^TX_0}_{0}\hat{\beta}_0 \\
&= 0
\end{aligned}$$

Expected values

$Se_0^2 = \frac{\|y - X_0\hat{\beta}_0\|^2}{n-p_0} = \frac{SSE_R}{n-p_0}$

under H_0 $\frac{(n-p_0) Se_0^2}{6^2} \sim \chi^2_{n-p_0}$

If H_0 is true, then

$$E\left[\frac{1}{n-p_0}\left\|\boldsymbol{y} - \boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0\right\|^2\right] = \sigma^2,$$

$E\left(\frac{(n-p_0) Se_0^2}{6^2}\right) = n-p_0$

Se_0^2

where $p_0 = p - k$.

If H_0 is true, then so is the full regression model $\boldsymbol{\eta} = \boldsymbol{X\beta}$, and so

Se^2

$$E\left[\frac{1}{n-p}\left\|\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}\right\|^2\right] = \sigma^2,$$

Hence what is

$\frac{SSE_D}{p-p_0}$

$$E\left[\frac{1}{p-p_0}\left\|\boldsymbol{X}\widehat{\boldsymbol{\beta}} - \boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0\right\|^2\right]?$$

$= 6^2$

$SSE_R = SSE_F + SSE_D \Rightarrow E(SSE_R) = E(SSE_F) + E(SSE_D)$

$>$

$6^2(n-p_0) = 6^2(n-p) + (p-p_0)6^2$

Null not correct

If the full model is correct, but H_0 is not, then it can be shown that

$$E\left[\frac{1}{p-p_0}\left\|\boldsymbol{X}\widehat{\boldsymbol{\beta}}-\boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0\right\|^2\right]>\sigma^2$$

Test statistic

under

Hence we can test H_0 by calculating

SSED / p-p₀ → σ² under H₀, > σ² under H_a

$$F = \frac{\left\|\boldsymbol{X}\widehat{\boldsymbol{\beta}} - \boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0\right\|^2 / (p - p_0)}{\left\|\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}\right\|^2 / (n - p)}$$

S_e^2

and rejecting it if F is ‘large’.

$SSE_F / n-p \rightarrow \sigma^2$

Theorem 12

任一 $\beta \neq 0$

Suppose $\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{\epsilon}$ with $\epsilon_i \sim N(0, \sigma^2)$ independently for $i = 1,2, \dots, n$. If $H_0: \boldsymbol{\eta} = \boldsymbol{X}_0\boldsymbol{\beta}_0$ is true, then

reduced

H_a: $\eta = X\beta$ full model

$$F = \frac{\left(\left\|\boldsymbol{X}\widehat{\boldsymbol{\beta}} - \boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0\right\|^2 / (p - p_0)\right)}{\left\|(\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}\right\|^2 / (n - p))} \sim F_{p-p_0, n-p}.$$

$$Se^2 = \frac{\|y - X\hat{\beta}\|^2}{n-p} \qquad \frac{(n-p)\,Se^2}{\sigma^2} \sim \chi^2_{n-p}$$

indep

$$Se_D^2 = \frac{\|X\hat{\beta} - X_0\hat{\beta}\|^2}{p-p_0} \qquad \frac{(p-p_0)\,Se_D^2}{\sigma^2} \sim \chi^2_{p-p_0}$$

$$F = \frac{\dfrac{(p-p_0)Se_D^2}{(p-p_0)\sigma^2}}{\dfrac{(n-p)Se^2}{(n-p)\sigma^2}} = \frac{Se_D^2}{Se^2} \sim F(p-p_0, n-p)$$

we reject H_0 if $F \geq F(p-p_0, n-p, \alpha)$

# of tested para

ANOVA table

MSE_D / Se_D^2

Source	SS	df	MS	F
SSR $\quad H_0$ vs M	$SSE_D \quad Q_0 - Q$	$p - p_0$	$\dfrac{Q_0 - Q}{p - p_0}\ (*)$	$F = \dfrac{*}{\dagger}$
SSE $\quad$ Error	$SSE_F \quad Q$	$n - p$	$\dfrac{Q}{n - p}\ (\dagger)$ $\rightarrow MSE_F$ Se^2	
SST $\quad$ Total	$SSE_R \quad Q_0$	$n - p_0$		

$\dfrac{Se_D^2}{So^2}$

where $\quad SSE_R = SSE_F + SSE_D$

$$Q = \left\|\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}\right\|^2 \text{ and } Q_0 = \left\|\boldsymbol{y} - \boldsymbol{X}_0\widehat{\boldsymbol{\beta}}_0\right\|^2.$$

and

$$H_0\!:\boldsymbol{\eta} = \boldsymbol{X}_0\boldsymbol{\beta}_0$$
$$M\!:\boldsymbol{\eta} = \boldsymbol{X}\boldsymbol{\beta}$$

$\beta_j = 0$

$s_4 - \beta_j \neq 0$

Remarks:

① P-value can also be calculated for this test and are provided by most statistical packages.

② This F-test may appears to be a one-sided test, but it is actually sensitive to any departure from

$$H_0: \mu_p = \mu_{p-1} = \dots = \mu_{p-k+1} = 0$$

③ When testing a single parameter, we can use either a test or an F-test. It can be proved that they are the same in this case.

Full $\quad Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

reduced $\quad Y_i = \beta_0 + \beta_1 X_1 + \varepsilon$

$X_0\beta_0 = y$

H_0: reduced

$\beta_2 = 0$

H_a: $X\beta = y$

Full model

$\beta_2 \neq 0$

Example 3.7

The price (y) in $\$1000\text{AUD}$, age ($x_1$) in decades, and area ($x_2$) in $1000\ m^2$, of five randomly chosen houses in a regional South Australian town are shown in the table below.

y	100	80	104	94	130
x_1	1	5	5	10	20
x_2	1	1	2	2	3

a) Fit a multiple linear regression model

$$y = \beta_0 + \beta_1 x_1 + \beta x_2 + \epsilon$$

to this data.

$y = 66.1252 - 0.3794 X_1 + 21.4365 X_2 + \varepsilon$

b) Test the hypothesis $H_0: \beta_1 = \beta_2 = 0$ against H_1: at least one of the $\beta_i \neq 0, i = 1,2$. Use $\alpha = 0.05$.

$H_0: y = \beta_0$ (reduced) $\quad H_a: y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ (Full)

$\beta_1 = \beta_2 = 0 \quad \beta_i \neq 0 \quad i = 1, 2.$

Example 3.7 Solution

a) $\hat{\beta} = \underbrace{(X^T X)^{-1}}_{3\times 3} \underbrace{X^T y}_{3\times 1}$

$$Y = \begin{pmatrix} \\ \\ \\ \end{pmatrix}_{5\times 1} \qquad X = \begin{array}{c} \begin{array}{ccc} 1 & X_1 & X_2 \end{array} \\ \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 5 & 2 \\ 1 & 10 & 3 \\ 1 & 20 & 3 \end{pmatrix}_{5\times 3} \end{array}$$

$X^T X = \begin{bmatrix} \\ \\ \end{bmatrix}_{3\times \cancel{5} \times 5\times 3}$ $\qquad (X^T X)^{-1} \qquad X^T y_{3\times 1}$

$$\hat{\beta} = \begin{pmatrix} 66.1252 \\ -0.3794 \\ 21.4365 \end{pmatrix}$$

Example 3.7 Solution

b)

To find the SST, we need to fit the reduced model first.

The reduced model is $\boldsymbol{Y} = \boldsymbol{x}_0 \boldsymbol{\beta}_0 + \boldsymbol{\epsilon}$, where

$$\boldsymbol{x}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } \boldsymbol{\beta}_0 = [\beta_0].$$

$$\beta_0 = (X_0^T X_0)^{-1} X_0^T Y$$

Observe that $\boldsymbol{x}_0^\top \boldsymbol{x}_0 = 5$, so $(\boldsymbol{x}_0^\top \boldsymbol{x}_0)^{-1} = 1/5$.

Hence, $\widehat{\boldsymbol{\beta}}_0 = (\boldsymbol{x}_0^\top \boldsymbol{x}_0)^{-1} \boldsymbol{x}_0 \boldsymbol{y} = \frac{1}{5} \sum_{i=1}^{5} y_i = \bar{y} = 101.60.$

If follows that $\boldsymbol{y} - \boldsymbol{x}_0 \widehat{\boldsymbol{\beta}}_0 = \boldsymbol{y} - \bar{y} = \begin{bmatrix} 100 \\ 80 \\ 104 \\ 194 \\ 130 \end{bmatrix} - \begin{bmatrix} 101.60 \\ 101.60 \\ 101.60 \\ 101.60 \\ 101.60 \end{bmatrix}.$

Hence, $SSE_R = \left\| \boldsymbol{y} - \boldsymbol{x}_0 \widehat{\boldsymbol{\beta}}_0 \right\|^2 = \left(\boldsymbol{y} - \boldsymbol{x}_0 \widehat{\boldsymbol{\beta}}_0 \right)^\top \left(\boldsymbol{y} - \boldsymbol{x}_0 \widehat{\boldsymbol{\beta}}_0 \right) = 1339.20.$

$n=5 \quad p=3 \quad p_0=1$

Example 3.7 Solution

$$F_{obs} = \frac{\underline{Se_D^2}}{Se^2} = \frac{\overbrace{\|X\hat{\beta} - X\hat{\beta}_0\|^2}^{SSE_D} / \underbrace{p-p_0}_{=2}}{\underbrace{\|y - X\hat{\beta}\|^2}_{SSE_F} / n-p \to 2} = \frac{478.2}{191.2} = 2.5.$$

$SSE_F = \|y - X\hat{\beta}\|^2 = 382.7$

$SSE_R = \|y - X_0\hat{\beta}_0\|^2 = 1339.20$

$SSE_D = SSE_R - SSE_F = 1339.2 - 382.7 = 956.2$

if $F \geq F_{2,2,0.05} = 19.$ we reject H_0

$F_{obs} < 19,$ F is not in the RR, $\therefore$

$\therefore$ can't reject H_0